

# DSAI 490 Assignment 2: Date Generator

## 1. Problem Formulation

The assignment is formulated as a conditional generation problem. Each input contains four conditions: weekday, month, leap-year status, and decade. The model must generate any date that satisfies all conditions. Exact-match accuracy is not the right primary metric because many different dates can be correct for the same input.

## 2. Data Representation

The condition side is tokenized using categorical tokens for weekdays, months, leap flags, and decade. Two output representations are used. For the Seq2Seq and Transformer models, the date is generated as a character sequence. For the GAN and VAE models, the date is represented as three categorical variables: day of month, month, and year.

## 3. Implemented Models

| Model                       | Formulation                                                                                                                | Training Objective                                         |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Conditional GAN             | Generator maps condition + noise today/month/year categories; discriminator scores real vs generated condition-date pairs. | Adversarial BCE + supervised categorical stabilization.    |
| Seq2Seq LSTM                | Encoder reads condition tokens; decoder generates the output date character by character.                                  | Character-level cross entropy with teacher forcing.        |
| Transformer Encoder-Decoder | Encoder reads condition tokens; decoder autoregressively generates the date string.                                        | Character-level cross entropy with causal decoder masking. |
| Conditional VAE             | Encodes condition-date pairs into a latent Gaussian and decodes condition + latent sample into day/month/year.             | Categorical reconstruction loss + KL regularization.       |

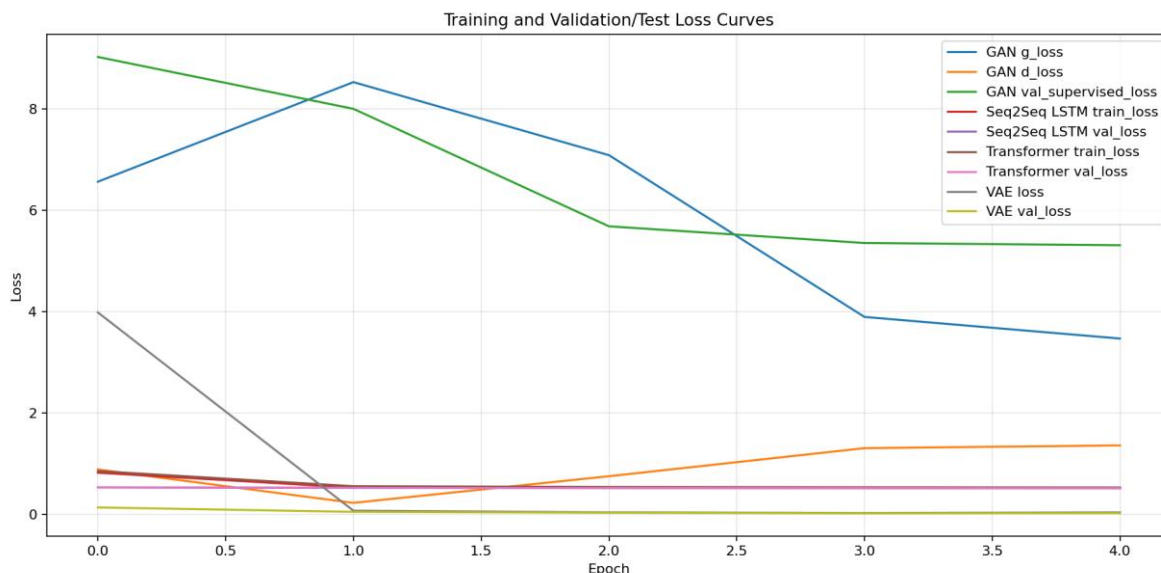
## 4. Training Setup

- The dataset is split into training and validation/test subsets.
- All models use a fixed random seed for reproducibility.
- Neural model checkpoints are saved after training.
- The notebook tracks training and validation/test losses for each model.

- Invalid neural samples are repaired by enumerating valid dates for the same condition; both raw and repaired behavior should be discussed if raw rates are reported.

## 5. Loss Curves

The combined training and validation/test loss curves generated by the notebook are shown below.



## 6. Output Analysis

The final prediction files were validated using the assignment constraints. A generated date passes only if its weekday, month, leap-year status, and decade match the input condition.

| Model        | Prediction Lines | Constraint Pass Rate | Output File                              |
|--------------|------------------|----------------------|------------------------------------------|
| GAN          | 14,647           | 100.00%              | predictions/gan_predictions.txt          |
| Seq2Seq LSTM | 14,647           | 100.00%              | predictions/seq2seq_lstm_predictions.txt |
| Transformer  | 14,647           | 100.00%              | predictions/transformer_predictions.txt  |
| VAE          | 14,647           | 100.00%              | predictions/vae_predictions.txt          |

All generated final prediction files passed the constraint validator after the notebook's repair step. This is suitable for final-format output submission, while the raw model outputs remain useful for comparing model quality.

## 7. Example Outputs

- [WED] [JAN] [False] [180] 1-1-1800
- [MON] [JAN] [False] [190] 1-1-1900
- [SAT] [JAN] [True] [200] 1-1-2000

## 8. Reflection

The most important modeling choice was to evaluate generated outputs by constraint satisfaction rather than exact match. The sequence models directly learn the date string format, while the GAN and VAE learn a structured categorical date representation. The deterministic validator/repair step guarantees valid final outputs and also makes model failures interpretable: when a raw output fails, the validator identifies that the generated date violated at least one of the four input constraints.

## 9. Submission Artifacts

- Notebook: DSAI\_490\_Assignment2\_Four\_Generative\_Models.ipynb
- Inference script: model/predict.py
- Weights: weights/\*.pt or model/weights/\*.pt depending on the final folder organization
- Prediction files: predictions/\*.txt
- Loss graph: training\_validation\_loss\_curves.png or figures/training\_validation\_loss\_curves.png